

Rational Exponents and Radical Expressions

Algebra Worksheet · Grade 8–10

Name: _____

Date: _____

Learning Objectives

- Convert expressions with rational exponents into radical form and vice versa
- Apply the laws of exponents (product rule, power rule, quotient rule) to simplify expressions with rational exponents
- Simplify radical and rational exponent expressions involving real numbers and variables

Problems

1. Rewrite the expression below in radical form:

$$x^{\frac{1}{2}}$$

2. Rewrite the expression below in radical form:

$$x^{\frac{1}{3}}$$

3. Convert the expression below from rational exponent form to radical form:

$$x^{\frac{3}{5}}$$

4. Evaluate the expression below by first converting it to radical form:

$$16^{\frac{1}{2}}$$

5. Evaluate the expression below by converting it to radical form:

$$27^{\frac{1}{3}}$$

6. Rewrite the radical expression below using a rational exponent:



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$$\sqrt[4]{x^3}$$

7. Use the product rule for exponents to simplify the expression below:

$$x^{\frac{1}{2}} \cdot x^{\frac{1}{3}}$$

8. Simplify the expression below using the laws of exponents:

$$x^{\frac{2}{3}} \cdot x^{\frac{1}{2}}$$

9. Use the power rule to simplify the expression below, then convert to radical form:

$$\left(x^{\frac{2}{3}}\right)^3$$

10. Simplify the expression below using the quotient rule for exponents:

$$\frac{x^{\frac{3}{4}}}{x^{\frac{1}{4}}}$$



Rational Exponents and Radical Expressions — Answer Key

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Answer Key

1. Answer: Square root of x

- The denominator of the rational exponent becomes the index of the radical.
- The numerator becomes the power of the base inside the radical.
- $x^{(1/2)} = \sqrt{x}$

2. Answer: Cube root of x

- The denominator 3 becomes the index of the radical.
- The numerator 1 means x is raised to the first power inside.
- $x^{(1/3)} = \sqrt[3]{x}$

3. Answer: Fifth root of x cubed

- The denominator 5 becomes the index of the radical.
- The numerator 3 is the power of x inside the radical.
- $x^{(3/5)} = \sqrt[5]{x^3}$

4. Answer: 4

- $16^{(1/2)}$ is the same as the square root of 16.
- The square root of 16 = 4.
- Answer: 4

5. Answer: 3

- $27^{(1/3)}$ is the same as the cube root of 27.
- The cube root of 27 = 3 because $3 \times 3 \times 3 = 27$.
- Answer: 3

6. Answer: x raised to the power 3/4

- The index of the radical (4) becomes the denominator of the rational exponent.
- The power inside the radical (3) becomes the numerator.
- $\sqrt[4]{x^3} = x^{(3/4)}$

7. Answer: x raised to the power 5/6

- Apply the product rule: add the exponents.
- $1/2 + 1/3 = 3/6 + 2/6 = 5/6$.
- $x^{(1/2)}$ times $x^{(1/3)} = x^{(5/6)}$

8. Answer: x raised to the power 7/6

- Apply the product rule: add the exponents.



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- $\frac{2}{3} + \frac{1}{2} = \frac{4}{6} + \frac{3}{6} = \frac{7}{6}$.
 - $x^{\frac{2}{3}}$ times $x^{\frac{1}{2}} = x^{\frac{7}{6}}$
-

9. Answer: x squared

- Apply the power rule: multiply the exponents.
 - $(\frac{2}{3})$ times 3 = $\frac{6}{3} = 2$.
 - Result: x^2
-

10. Answer: x raised to the power 1/2, which equals the square root of x

- Apply the quotient rule: subtract the exponents.
 - $\frac{3}{4} - \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$.
 - $x^{\frac{3}{4}}$ divided by $x^{\frac{1}{4}} = x^{\frac{1}{2}} = \sqrt{x}$
-

